Manuscript outline: what each paragraph should contain

enceladus_DSM `studies/keff_sintering/manuscript_outline.tex`

2026-10-12, against the draft of the same date

Framing. Dry snow metamorphism is the story; the model covers one part of it, sintering by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach. Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to $-5\,^{\circ}\mathrm{C}$, air at 1 atm, sealed pores, 50 μm grains, 30 days, two dimensions.

How to read this. This is a plan of content: each numbered item is one paragraph and lists what it should say, with the numbers. Line numbers (L first–last) are those of the draft of 2026-10-12 in `Literature/` (36 pages, 669 lines). **[written]** the paragraph exists and says this; **[to write]** it does not exist yet; **[fix]** one specific thing in it is wrong; **[trim]** it can be shorter. Boxes mark where a figure or table goes.

Sections 4 and 5 of the draft are treated as notes. They were written for an earlier model and an earlier set of results. Nothing they say about results is carried over. They are used only as a source of points about the literature, marked **[from the notes]** , each to be checked against its source.

Suggested citations follow each paragraph, with the key in your `.bib`. Most are the ones the old Discussion used. I have matched them by title only; check each claim against the paper before it goes in.

Decisions already made, which this plan follows

Collected from the work of 2026-09-08 to 2026-10-12: the session log, the campaign record and the manuscript plan.

- **Framing** (10-08, 10-10). Laboratory-observable sintering by vapor transport, inside the dry-snow-metamorphism story. The Introduction and title stay as written; scope caveats go in the Methods. No timescales for Enceladus temperatures; the timescale map is dropped. Advisor's comments still to come.
- **Comparisons** (10-08). Not natural snow (angular, more porous, loaded). The analogues are unloaded ice-sphere and ice-powder samples: Molaro et al., Demmenie et al., Choukroun et al. From the snow literature only functional forms carry over, never magnitudes.
- **Kinetics** (09-22, 10-09). α_c constant in temperature; the thin-interface term always on; aggregates stated as $\alpha_c = 1.020 \times 10^{-3}$. No further simulations for this paper.
- **Two-grain argument** (09-09, 10-02, 10-13). Rests on what the vapor route does and does not reproduce, and on exponents; no Thomas or Kingery runs. The two mismatches have different causes and neither is attributed to missing surface diffusion (Appendix, this outline): the rate at

$-20\,^{\circ}\mathrm{C}$ is a matter of how fast vapor can be supplied; the exponent at saturation is a matter of resolving a small neck with a diffuse interface. Molaro's conventions: neck *width*, time anchored on the first measurement. The $-20\,^{\circ}\mathrm{C}$ series is the benchmark.

- **Conductivity** (09-23). Tensor law in the interface; the cell problem is a steady problem on a frozen microstructure and says nothing about the transient model. Two dimensions only, so no absolute k_{eff} is claimed.
- **Clock and normalization** (10-08, 10-12). The clock is τ_{sub}; curves are normalized at $\theta_r = 30$. The first day depends on the interface parameter, so a "rise from day 1" is never quoted as a property of the ice.
- **Notation** (09-28 to 09-30). ϕ phase field, φ porosity, ζ indicator, χ curvature; subscripts "ice" and "air", not i and a; no index notation; a single dot for every contraction; the vector $\mathbf{t}$ named as in Auriault et al.
- **Citations** (09-23). Only work whose setting matches ours; every DOI checked against Crossref.
- **Words.** Desublimation ("deposition" is plume fallout); "vapor"; ε is the interface parameter, not the interface width.

Where the draft stands

Section	State
Key points, Abstract, Summary	to write (placeholders)
1 Introduction	written; small fixes and trims
2.1–2.3 Model, kinetics, homogenization	written; four definitions to add
2.4, 2.5 Numerics, simulation design	to write (material exists in Section 3)
3 Grain pairs	written, both pairs; numbers and one correction to add
4 Aggregates	written; numbers to add, one paragraph to correct
5 Discussion	to write; plan below
6 Conclusions	to write

New in this draft: the temperature table (Table 2) is in and the table notes resolve; the saturated-pair paragraph is written (L339–353); the Discussion has a new paragraph on the Molaro comparison (L436–453).

Section titles

Descriptive headings in place of "Methods" and "Results"; the Results become two sections. The rest of this outline uses the new titles.

Draft	Suggested
1 Introduction	1 Introduction
2 Methods	2 Modeling Approach
2.1 Phase-Field Model For Dry Snow Metamorphism	2.1 Phase-Field Model of Dry Snow Metamorphism
2.2 Gibbs-Thomson Condition	2.2 Interface Kinetics and the Sintering Time Scale
2.3 Homogenizing for Effective Thermal Conductivity	2.3 Effective Thermal Conductivity by Homogenization
–	2.4 Numerical Implementation
–	2.5 Simulation Design: Grain Pairs and Aggregates
3 Results	–
3.1 Phase-Field Model Validation	3 Neck Growth Between Two Ice Grains
	3.1 Comparison With Unsaturated Experiments
	3.2 Growth at Saturation
3.2 Aggregate Sintering and Effective Thermal Conductivity	4 Thermal Conductivity of Sintering Aggregates
	4.1 Microstructural Evolution
	4.2 Temperature Rescales Time
	4.3 Conductivity as a Function of Specific Surface Area
4 Discussion	5 Discussion
4.1 What vapor transport alone reproduces	5.1 What Vapor Transport Alone Reproduces
4.2 SSA is a universal predictor of $\mathbf{K}_{\text{eff}}$	5.2 Temperature as a Rescaling of Time
	5.3 Specific Surface Area as the State Variable
	5.4 What a Laboratory Could Measure
	5.5 Implications for Plume Deposits on Enceladus
	5.6 Limitations
5 Conclusion	6 Conclusions

"SSA is a universal predictor" claims more than the result: the power law holds for $\varphi \leq 0.375$ and porosity sets the level.

Build check (LaTeX warnings)

From the Overleaf project uploaded on 2026-10-11 (the newer draft came as a PDF only). As uploaded: **561 overfull-box warnings**, 144 underfull, 19 undefined citation keys, 8 undefined references. With the changes below, tested on a copy: **0 overfull, 0 underfull, 0 undefined citations**; the undefined references remain because their targets do not exist.

1. **554 of the overfull warnings: every line of body text.** The class sets ragged-right text with a 24 pt margin and only 2 em of slack (`agujournal2019.cls`, lines 113–115). TeX cannot break paragraphs within that slack, gives up, and lets each line run into the margin. Immediately after `\begin{document}`:

   ```
   \setlength{\RaggedRightRightskip}{0pt plus 4em}\RaggedRight
   \hbadness=10000 \vbadness=10000
   ```

   The first line removes the overfull warnings and leaves the page count where it is (37 pages against 38). The second hides the underfull warnings, which come from figure captions and page breaks and are harmless. Keeping the class's 24 pt margin (`24pt plus 4em`) also works and costs two pages.

2. **The other 7: the figures.** Each is included at 170 mm; the text block is 5.5 in (139.7 mm), so each sticks out 86 pt to the right. To keep the true size and center it on the text block:

   ```
   \makebox[\textwidth][c]{\includegraphics[width=170mm]{...}}
   ```

   or use `width=\textwidth` in the draft and supply the 170 mm files at submission.

3. **Table 2 was taller than the page** by 168 pt ("Float too large"). Cause: in draft mode the class gives `\footnotesize` a 20 pt line spacing, so the tables stayed double-spaced. Fixed in `tables.tex` (2026-10-11): the tables set their size with `\fontsize{8}{10}`. Table 2 now takes half a page. Re-upload `tables.tex`; nothing changes in the manuscript source.

4. **19 undefined citation keys**, all in Sections 4–5. The text uses an older key style; every one has an entry in the `.bib` under a new key. Find and replace:

`bouvet_snow_2022`	`bouvetSnowEquiTemperatureMetamorphism2022`
`braun_rigorous_2024`	`braunRigorousApproachSpecific2024`
`byrne_radar_2024`	`byrneRadarAttenuationSubsurface2024`
`calonne_numerical_2011`	`calonneNumericalExperimentalInvestigations2011`
`calonne_study_2014`	`calonneStudyTemperatureGradient2014`
`calonne_thermal_2019`	`calonneThermalConductivitySnow2019`
`fourteau_impact_2021`	`fourteauImpactWaterVapor2021`
`gallet_measurement_2009`	`galletMeasurementSpecificSurface2009`
`granger_orientation_2021`	`grangerOrientationSelectiveGrain2021`
`jaumann_distribution_2008`	`jaumannDistributionIcyParticles2008`
`kaempfer_phase-field_2009`	`kaempferPhasefieldModelingDry2009`
`libbrecht_measurements_2013`	`libbrechtMeasurementsSurfaceAttachment2013`
`libbrecht_physical_2017`	`libbrechtPhysicalDynamicsIce2017`
`matzl_measuring_2006`	`matzlMeasuringSpecificSurface2006`
`molaro_microstructural_2019`	`molaroMicrostructuralEvolutionWater2019`
`proksch_density_2015`	`prokschDensitySpecificSurface2015`
`schneebeli_tomography_2004`	`schneebeliTomographyTemperatureGradient2004`
`spencer_enceladus_2009`	`spencerEnceladusActiveCryovolcanic2009`
`spencer_enceladus_2013`	`spencerEnceladusActiveIce2013`

   Check `fourteau_impact_2021`: the `.bib` also has `fourteauMacroscopicWaterVapor2021`.

5. **Undefined references.** The table notes are resolved in the new draft. Left: six in the old Discussion, to figures and sections that no longer exist; they go when that text is replaced.

6. **Unused files in the project.** `Figure8_timescales.pdf` (dropped); `Figure7_state_law_allphi.pdf`; `agujournaltemplate.tex`; `si_template_2019.tex` (needed later for the supplement).

Front matter

- **Title.** As is, without "Version 2".
- [to write] **Key points** (three, each at most 140 characters).
  1. For a given ice-grain packing, temperature from −40 to −5 °C only rescales the time over which sintering raises conductivity.
  2. Effective conductivity follows specific surface area by one power law in well-connected packings; porosity sets its level.
  3. Vapor transport alone reproduces part of the neck growth measured between ice grains in the laboratory.

- [to write] **Abstract** (at most 250 words). Sentence by sentence:
  1. Setting: porous ice on icy bodies and on Earth; its conductivity depends on microstructure, which sintering changes.
  2. Gap: how conductivity evolves as grains sinter without compaction, and what temperature and porosity do to it.
  3. Method: a phase-field model of sublimation and desublimation with vapor and heat transport; conductivity by periodic homogenization.
  4. Grain pairs: against Molaro et al., the vapor route gives 41% of the measured neck growth at $-20\,^{\circ}\mathrm{C}$; at saturation the neck grows as $t^{0.21}$.
  5. Aggregates: 125 simulations, five porosities, five temperatures, 30 days.
  6. Result 1: for a given packing, conductivity against specific surface area is the same at every temperature within 0.3%; temperature sets the rate.
  7. Result 2: for porosity up to 0.375, conductivity scales as $\mathrm{SSA}^{-0.80}$; porosity sets the level.
  8. Meaning: SSA, which can be measured, tracks the conductivity of a sintering deposit; what a laboratory test would look like.
- [to write] **Plain Language Summary** (at most 200 words). The first half of the draft's summary (L33–43) can stay. Then: what was simulated; temperature changes how fast, not what happens; surface area tells you the conductivity; why it matters for plume deposits.

1 Introduction

Kept as drafted; no reordering. Each paragraph and what it carries:

1. **What dry snow metamorphism is** (L57–68). [written]

   Message: *Ice near its melting point rearranges itself to reduce surface area, and that changes its bulk properties.*

   Definition; everyday examples; that it changes bulk properties. [trim] The examples return in paragraph 2; keep them in one.

2. **Why it matters on Earth** (L69–81). [written]

   Message: *This process controls how snow and firn carry heat and load, so it has been studied closely on Earth.*

   Avalanches, firn, conductivity of snowpacks.

3. **Icy bodies and Enceladus** (L82–101). [written]

   Message: *The same process should act on plume deposits near Enceladus's fractures, where its effect on conductivity is unknown.*

   Plume, grain size, where sintering is expected, strength changes. [trim] Last sentence (L99–101) is circular. [fix] "inhabit" → harbor.

4. **Sintering as part of metamorphism** (L102–116). [written]

   Message: *The first stage is neck growth between touching grains, by vapor transport and surface diffusion; two laboratory studies of grain pairs give us something to compare with.*

   Neck growth, the two routes, earlier work, Molaro, Choukroun. [fix] "deposits" → desublimates. Add one clause: Demmenie et al. is the second comparison used here.

Figure 1 [written] placed. [fix] Caption and figure say "neck of radius $R(t)$"; the paper uses the neck width w.

5. **Attachment kinetics** (L117–134). [written]

   Message: *How fast vapor attaches to ice is set by a coefficient that is known only to within two decades; we hold it constant.*

   Supersaturation; the Wilson–Frenkel law; α_c. [trim] [fix] Shorten to those three, then end on: α_c is uncertain over two decades (10^{-3}–10^{-1}) and is held constant within each study here. As drafted the paragraph builds toward a temperature-dependent α_c through $\sigma_0(T)$, which the model does not have.

6. **Metamorphism beyond a pair; earlier models** (L135–154). [written]

   Message: *Phase-field models can follow a whole microstructure, but so far each has been tuned to one data set.*

   Kaempfer & Plapp; Bouvet; Braun; that they tune α_c. [trim] The first three sentences repeat paragraph 4.

7. **What is not known** (L155–166). [written]

   Message: *Nobody has shown how conductivity evolves as an ice aggregate sinters without compacting, or how temperature and porosity enter.*

   Bulk properties without compaction; the roles of temperature and porosity. [trim] [fix] Drop the claim that no model handles varying temperature and saturation (this paper does not close that gap) and the second telling of Bouvet and Braun.

8. **What follows** (L167–180). [written]

   Message: *We compare the model with grain-pair experiments, then use it on aggregates to relate conductivity to microstructure.*

   Order of the paper. [fix] 30-day, not 28; "independent of"; delete the fragment at the end; add the saturated pair; say "compare with", not "validate"; the predictor is independent of temperature, and of porosity only in its exponent.

2 Modeling Approach

Draft: 2 Methods.

Opening (L182–189). [written] [trim] Three sentences: the model descends from Kaempfer & Plapp and Moure & Fu; what is new here is its application, not the numerical method; what the section covers.

2.1 Phase-Field Model of Dry Snow Metamorphism

1. **Unknowns** (L191–205). [written]

   Message: *The model tracks where the ice is, the temperature and the vapor density.*

   Two paragraphs; the first (what a phase field is) can be one sentence. Fields ϕ_i, ϕ_a (air), T, ρ_v.

2. **Scope.** (new) [to write]

Message: *The model contains one sintering route, vapor transport, and the reader should know that before reading the results.*

Three or four sentences. Of dry snow metamorphism the model carries sintering by vapor transport: sublimation, diffusion through the pore, desublimation. Not in it: surface diffusion, grain-boundary processes, densification, overburden. Define desublimation. The laboratory analogues are therefore unloaded ice-sphere samples, not natural snow.

3. **Phase equation** (L206–212). **[written]**

   Message: *Ice grows or shrinks according to curvature and to how far the local vapor is from saturation.*

   The two terms of Eq. (1a). **[to write]** Add: M_{sub} and α_{sub} in terms of τ_{sub} and ε (or a pointer to Moure & Fu); ε named the interface parameter, with the visible band $9.2\,\varepsilon$.

4. **Saturation vapor density** (L213–217). **[written]**

   Message: *Saturation depends strongly on temperature; this is where nearly all the temperature dependence of the model comes from.*

   [fix] Check the last term of Eq. (3) against the source ($k_5 \ln(T + T_0)$?). One attribution: the text cites Flatau and Wexler, Table 1 says ASHRAE.

5. **Energy equation** (L218–224). **[written]**

   Message: *Heat is conducted and released or absorbed at the interface.*

   Mixture rules; constant properties.

6. **Vapor equation** (L225–231). **[written]**

   Message: *Vapor diffuses through the pore and is exchanged with the ice.*

   Diffusivity. **[to write]** Add one sentence on the temporal scaling ξ_T, ξ_v that Tables 3 and 4 list.

Table 1 **[written]** Material properties. Cited at L230.

2.2 Interface Kinetics and the Sintering Time Scale

Draft: 2.2 Gibbs-Thomson Condition.

1. **The interface condition** (L232–242). **[written]**

   Message: *In the sharp-interface limit the model obeys the Gibbs–Thomson condition, with a kinetic coefficient fixed by the condensation coefficient.*

   Eqs. (6), (7). **[to write]** Add $d_0 = \gamma m/(\rho_{\mathrm{ice}} k_B T)$ in the text.

2. **The thin-interface limit.** (new) **[to write]**

   Message: *A diffuse interface adds a known amount to the kinetic coefficient; we include it and say how large it is.*

   The model recovers Eq. (6) with a kinetic coefficient $\beta_{\mathrm{sub}} + \Delta\beta$, where $\Delta\beta = a_1 a_2\, \varepsilon\, (D_T^{-1} + D_v^{-1})\, \rho_{\mathrm{ice}}/\rho_{vs}^I$ (Kaempfer & Plapp, 2009; Moure & Fu, 2024). It is 2% of β_{sub} in the aggregates and 23% in the Molaro pairs.

   Cite: Kaempfer & Plapp 2009 `kaempferPhasefieldModelingDry2009`; Moure & Fu 2024 `mourePhaseFieldModelWet2024`.

3. **The time scale.** (new) [to write]

   Message: *The model has one characteristic time, and time divided by it is the "age" used to compare temperatures.*

   $\tau_{\text{sub}} = \varepsilon^2(\beta_{\text{sub}} + \Delta\beta)/d_0$; the sintering age $\theta = t/\tau_{\text{sub}}$. ε is the same in every aggregate run, so θ compares temperatures on one clock. The 30-day runs end at $\theta = 43$ ($-40\,^\circ$C) to 1256 ($-5\,^\circ$C). The clock is τ_{sub} (decided 2026-10-12). Be explicit that there are two time scales of the same form, $\ell^2\beta/d_0$, the time for a surface of curvature $1/\ell$ to move a distance ℓ by sublimation and desublimation: τ_{sub} with $\ell = \varepsilon$, the relaxation time of the diffuse interface, which is the parameter the model is written with; and $\tau_{\bar{R}} = \bar{R}^2\beta/d_0$ with $\ell = \bar{R}$, the time for a grain to change shape, which is what the scaling analysis produces. Neither contains the other; $\tau_{\text{sub}} = (\varepsilon/\bar{R})^2\,\tau_{\bar{R}} = \tau_{\bar{R}}/2500$. Suggested names: "interface relaxation time" and "grain-scale sintering time". One sentence to keep it honest: τ_{sub} contains ε, so the values of θ quoted are for $\varepsilon = 1$ μm; ratios of τ_{sub} between temperatures do not depend on it.

4. **The values used.** (new) [to write]

   Message: *The condensation coefficient is a choice, constant in temperature, and the reader needs the three values in one place.*

   α_c is constant in temperature within each study: 1.020×10^{-3} (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair). Why constant: a temperature-dependent α_c would make a temperature sweep uninterpretable.

   Cite: range of α_c: Braun et al. 2024 `braunRigorousApproachSpecific2024`; Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Libbrecht 2017 `libbrechtPhysicalDynamicsIce2017`.

Table 2 [written] Temperature-dependent quantities. [fix] Not cited in the text; cite it in paragraph 3.

2.3 Effective Thermal Conductivity by Homogenization

1. **Two scales and the microscale problem** (L244–254). [written]

   Message: *If grains are much smaller than the sample, the sample conducts like a uniform material with an effective conductivity.*

   [fix] Periodicity is an assumption ("without loss of generality" is not true of it).

2. **Macroscale law and cell problem** (L255–263). [written]

   Message: *That effective conductivity comes from solving one auxiliary problem on a periodic cell.*

   [fix] Finish the broken sentence ("referred to as co…"); Eq. (11d) needs $\nabla\mathbf{t}$.

3. **Single-domain form** (L264–276). [written]

   Message: *The two-phase problem can be written as one problem over the whole cell.*

   [trim] Eqs. (12)–(14) can be one sentence leading to Eq. (15).

4. **Diffuse form and the tensor law** (L277–291). [written]

   Message: *With a diffuse interface the conductivity inside the interface must be direction-dependent, or the answer is biased; we use the law that removes the bias.*

   Eq. (16), (17); why the tensor law (first-order error removed). [to write] Add two sentences. Credit: Nicoli, Plapp & Henry (2011) define the law and show the cancellation for a planar

interface at contrasts of 2 and 10; here it is applied to the cell problem at a contrast of 115 and verified on laminate and disk-array cases (supplement). Why it matters here: with a scalar law the bias reaches tens of percent and grows with $\varepsilon\,\mathrm{SSA}$, so it would drift as the aggregate sinters. State that the cell problem is solved on frozen snapshots of ϕ.

Cite: Nicoli, Plapp & Henry 2011 `nicoli_tensorial_2011`; Auriault 1991 `auriaultHeterogeneousMediumEquivalent1991`; Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

5. **What is reported.** (new) **[to write]**

   Message: *Define the two quantities every results figure plots: the scalar conductivity and the specific surface area.*

   Three definitions: $\mathbf{K}_{\mathrm{eff}}$ in the diffuse form, the one evaluated; $k_{\mathrm{eff}} = (k_{xx} + k_{yy})/2$; and SSA, how it is measured from ϕ and its units. SSA is used throughout Section 4 and is defined nowhere.

   Cite: for computing SSA from a field, possibly Flin et al. 2014 `flinComputationsSpecificSurface2014` (check that the method matches).

Figure 2 **[written]** placed. **[fix]** Not cited in the text. Caption: "red" lines are orange; add that only the x problem is shown.

2.4 Numerical Implementation

1. **Discretization and time stepping.** (new) **[to write]**

   Message: *Enough for someone to reproduce the runs.*

   Quadratic C^1 splines; element size $\varepsilon/\sqrt{2}$; adaptive step capped at $2\,\tau_{\mathrm{sub}}$; conductivity sampled every step for $\theta < 11$, then at every 0.1% drop in SSA.

2. **Checks.** (new) **[to write]**

   Message: *The results do not depend on domain size, time step or sampling; they depend on the interface parameter only during the first day.*

   One sentence each, pointing to the supplement: domain size (Fig. S1); interface parameter (Fig. S2), including that the first day depends on ε and later rates do not; time step and sampling (Fig. S3).

Table 3 **[written]** Complete parameters. **[fix]** Not cited in the text. *Open: keep it here, or the short version here and this one in the supplement.*

2.5 Simulation Design: Grain Pairs and Aggregates

1. **Grain pairs.** (new) **[to write]**

   Message: *What the three two-grain simulations are, so that Section 3 can go straight to results.*

   The material is in the draft at L315–327 and L339–347; move it here. Axisymmetric about the line of centers; the two Molaro geometries and the equal-grain pair ($R = 84.4$ μm, initial neck width 28 μm, $-20\,^{\circ}$C, 100 h); fixed temperature and humidity at the outer boundary; how the neck width w is measured.

Table 4 [written] Two-grain parameters. [to write] Add a column for the saturated pair. [fix] Simulated time at −5 °C is 68 min.

2. **Aggregates.** (new) [to write]

   Message: *What the 125 aggregate simulations are and how the packings were made.*

   The material is in the draft at L357–366; move it here. 2 mm square, periodic in both directions; circular grains, log-normal radii with mean 50 μm, placed by drop-and-roll deposition; five porosities, five independent packings each, the same packings at all five temperatures: 125 runs of 30 days; uniform temperature, vapor initially at saturation. Domain size: $L/\bar{R} = 40$, at the 1–2 mm representative volume of Calonne et al. (2011). Packings are accepted only if contacts across the periodic seam are as dense as in the interior; seeds are independent across porosity (details in the supplement).

   Cite: representative volume: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

3. **Why periodic, why two dimensions.** (new) [to write]

   Message: *Justify the two choices a reviewer will question first.*

   Periodic: the domain stands for the interior of a deposit, no wall physics has to be invented, and the system is closed, so curvature alone drives transport. Two dimensions: cost; three-dimensional effects are discussed in Section 5.6.

3 Neck Growth Between Two Ice Grains

Draft: 3 Results (opening) and 3.1 Phase-Field Model Validation.

Section opening (L293–308). [trim] Two sentences: before using the model on aggregates, we ask how much of measured neck growth it reproduces, with one experiment away from saturation and one at saturation. The rest of the drafted opening repeats the Introduction.

3.1 Comparison With Unsaturated Experiments

1. **Why these experiments** (L309–314). [written]

   Message: *They are the cleanest published measurements of neck growth between two ice grains at these sizes and temperatures.*

2. **What is fitted and what is not** (L315–327). [written] in part.

   Message: *One quantity is inferred from the data (the humidity, from grain shrinkage) and one is chosen (the condensation coefficient, at its upper limit); nothing is tuned to the neck.*

   Include: the chamber humidity is not reported, so it is inferred at each temperature from the measured shrinkage of the large grain (0.99715 at −20 °C; 0.99674 at −5 °C); $\alpha_c = 0.1$, the top of the literature range, chosen as the fastest plausible vapor route. [fix] "The only free parameter is β_{sub}": the humidity is fitted too.

3. **Result** (L328–338). [written] panels (a), (b) in words.

   Message: *Vapor transport alone gives less than half of the measured neck growth at −20 °C; the −5 °C comparison is not reliable.*

   [to write] Add: at −20 °C the model gives 41% of the measured neck growth over 78 min; at −5 °C it gives 170% over 48 min (52.1 against 44.0 μm); the authors' caveat on the −5 °C

series; panels (c), (d), with the statement that the shrinkage is what the humidity was fitted to and so is not a test. Use Molaro's conventions (neck width, time from the first measurement) and say so. One sentence of context: Molaro's own model matches these data through its vapor term, which assumes $\alpha_c = 1$ and no gas diffusion (their Appendix A); ours has $\alpha_c = 0.1$ and diffusion through air.

Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

Figure 3 [written] placed. [fix] Caption stops at (b); say what the colors in (a) show.

3.2 Growth at Saturation

1. **Set-up and result** (L339–353). [written]

   Message: *With the grains held at constant size, the neck grows as a power law in time with an exponent of about 0.2, below the measured and theoretical values; away from saturation the exponent drops, as observed.*

   Grain size, temperature, duration, how the saturation is set, the fit window, exponent 0.21. [to write] Add: why this run (at saturation the grains do not shrink, radius +0.06%, so neck growth is isolated); the neck goes from 28 to 63 μm; Demmenie et al. measure 0.26–0.33 at saturation and about 1/7 when undersaturated; the fitted exponent rises from 0.21 to 0.23 as the window starts later. For contrast, the same fit to the unsaturated $-20\,^{\circ}\mathrm{C}$ run of Section 3.1 gives 0.135, close to 1/7, so the model does reproduce the drop in exponent away from saturation; Molaro's own $-20\,^{\circ}\mathrm{C}$ data give 0.23 ± 0.03 (refitted 2026-10-12 after the relaxation period; `studies/molaro_2019/exponents/`). Only $-20\,^{\circ}\mathrm{C}$ is quoted: the $-5\,^{\circ}\mathrm{C}$ series is set aside (decided 2026-10-12).

   Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`.

   [fix] L343–344 says $\alpha_c = 10^{-3}$ puts growth in "a vapor-limited regime … governed by vapor transport". It is the reverse: at this α_c growth is limited by attachment (gas diffusion adds 0.4–2% to the resistance). That is also the regime the $t^{1/3}$ law describes, so the comparison stands.

2. **Why the exponent is low.** (new) [to write]

   Message: *The sharp-interface law that the model is built to obey predicts 0.29 for this very run. The run gives 0.21 because for its first tens of hours the neck is only a few interface widths across and grows too fast, which flattens the fitted curve. It is a resolution effect, not missing physics.*

   State the evidence in this order (numbers and method in the Appendix of this outline):

   - The Gibbs–Thomson prediction for this geometry and these parameters gives an exponent of 0.29 and the final neck width to 4%.
   - The run's neck velocity divided by that prediction is 2.5 when the neck fillet is 2 interface bands across, 1.2 at 5 bands and 1.0 at 7 bands: the excess follows the ratio of interface width to fillet radius.
   - The exponent fitted from later and later start times rises (0.215 from 2 h, 0.236 from 71 h).
   - The run starts from a neck already 28 μm wide, so the fit must guess how long that neck "had been growing". The free fit takes 4.8 h; the prediction says 16 h, and with that value the same data give 0.28. An earlier run of this model from grains just touching gave 0.283.

State the limit of the claim: within 100 h the velocity reaches the prediction but the fitted exponent does not (0.236 against 0.29), and the test that would settle it, a run at half the interface width, was not done. *Open: how firmly to word this. "Consistent with" is supported; "caused by" is not yet.*

Figure 4 [written] placed. [fix] Caption: add temperature, grain size and the fit window (1–100 h).

4 Thermal Conductivity of Sintering Aggregates

Draft: 3.2 Aggregate Sintering and Effective Thermal Conductivity.

4.1 Microstructural Evolution

1. **The gallery** (L355–372). [written]

   Message: *Show the reader what sintering does: contacts become necks, sharp features round off, and the grains do not move.*

   Already there: in the second half (L366–372): point contacts become necks, small grains and sharp features disappear, the pore space rounds. [to write] Add what the colors show: vapor is undersaturated over convex ice, which sublimates, and supersaturated in necks and concavities, where it desublimates. Move the set-up (first half) to Section 2.5.

Figure 5 [written] placed, with caption.

2. **The conductivity tensor** (L373–382). [written]

   Message: *The packings are close to isotropic at low porosity and not at high porosity, which matters later.*

   [trim] One sentence on how it is obtained (Section 2.3). Include: k_{xy} is zero in the mean over packings and up to 12% of k_{xx} on a single one; k_{xx}/k_{yy} is near 1 for $\varphi \leq 0.325$, 0.93 at 0.375, and about 0.8 and 0.65 at $\varphi = 0.425$ and 0.475, with y the direction of deposition; the scalar k_{eff}, the mean of the two, is used from here on.

4.2 Temperature Rescales Time

3. **Figure 6, one packing at five temperatures** (L383–391). [written]

   Message: *Temperature changes how fast a packing sinters and nothing else: plotted against age or against surface area, the five temperatures are one curve.*

   panels (b)–(d) in words. [to write] Add, in this order:

   - Panel (a): the packing ($\varphi = 0.325$) at four instants.
   - In time, warmer runs change faster. To give a size, use numbers that do not depend on the interface parameter: between $\theta_r = 30$ and $\theta = 331$, k_{eff} rises by a factor 1.171 ± 0.015 and SSA falls to 0.819 ± 0.007 of its value, for every packing with $\varphi \leq 0.375$. The rise from day 1 (7–12% at $-40\,^{\circ}\mathrm{C}$, 31–44% at $-5\,^{\circ}\mathrm{C}$) may be quoted only as specific to $\varepsilon = 1\ \mu\mathrm{m}$: doubling ε changes it by 9 points and leaves the later rate unchanged.

- Against θ the five curves fall on one: rates follow τ_{sub} within 4%.
- Against SSA they agree within 0.27% for every one of the 25 packings (median 0.07%). This is the result; state it as a sentence.
- SSA falls linearly in $\log\theta$, about 0.13 per decade, after the relaxation period: the form reported for isothermal snow (Legagneux et al.; to verify before citing).
  Cite: Legagneux & Dominé 2005 `legagneuxMeanFieldModel2005`; Taillandier et al. 2007 `taillandierRateDecreaseSpecific2007`. Both are in your `.bib`; for the functional form only.
- Why curves are normalized at $\theta_r = 30$ and not at $t = 0$: the first day depends on the interface parameter (Fig. S2).

Figure 6 [written] placed. [fix] Caption: $|\sigma| \leq 0.34$; define θ_r.

4.3 Conductivity as a Function of Specific Surface Area

4. **Figure 7, porosity** (L392–410). [written]

   Message: *Across packings, surface area gives the relative change in conductivity through one power law, and porosity sets the level; this holds up to a porosity of 0.375.*

   Already there: the order (a), (b), (c), (d) and the 0.3% statement. Include:

   - (a) At $-20\,^{\circ}\mathrm{C}$, mean of five packings and their range, for each porosity. At 30 d the conductivity differs by a factor of about 3.3 between $\varphi = 0.275$ and 0.475.
   - (b) Fifteen packings ($\varphi \leq 0.375$, five each) at four temperatures, 60 curves. $-40\,^{\circ}\mathrm{C}$ is absent because those runs end at $\theta = 43$, too soon after the reference age. All follow $k_{\mathrm{eff}}/k_{\mathrm{eff},r} = (\mathrm{SSA}/\mathrm{SSA}_r)^p$ with $p = -0.80$; by porosity -0.79, -0.82, -0.78.
   - (c) The level: $k_{\mathrm{eff},r}/k_{\mathrm{ice}} = 1.37\exp(-5.01\,\varphi)$ for $\varphi \leq 0.375$, with 6.6% scatter between packings. Exponential in porosity, not linear.
   - (d) Exponent of each packing. For $\varphi = 0.425$ and 0.475 it is weaker (-0.73, -0.63) and more scattered, and those packings are anisotropic; they are shown and not fitted.
   - What it amounts to: SSA gives the relative change in conductivity; porosity gives the level; neither alone gives both.
   - One closed form, if wanted: $k_{\mathrm{eff}} = 1.37\,k_{\mathrm{ice}}\,e^{-5.0\varphi}\,(\theta/30)^{0.066}$ for $\varphi \leq 0.375$ (fit of 2026-10-06).

   [fix] Three statements to correct: "3 packings for each porosity" (five); the unfinished sentence on connectivity; and the statement that the ice "remains disconnected in the horizontal direction". The last is not true of these packings: a percolation test on all 25 (2026-10-12, `studies/keff_sintering/percolation/`) finds the ice connected across the periodic cell in both directions in every packing from 10 d on, with 92–99% of the ice in the spanning cluster. Say instead what the data show (2026-10-12): the packings become anisotropic as porosity rises ($k_{xx}/k_{yy} = 1.02$, 1.03, 0.93, 0.82, 0.66, with y the direction of deposition), and the two directions then follow different laws. Along the deposition direction the exponent stays at -0.79 to -0.86 up to $\varphi = 0.425$ (-0.73 at 0.475); across it the exponent falls to -0.70, -0.60 and -0.48 from $\varphi = 0.375$ on. The isotropic mean is therefore not one power law at high porosity, and those packings are shown and not fitted. *Open: whether to plot or tabulate the directional exponents; whether to add that a porosity-dependent representative volume may contribute. The second is untested: the domain-size study was run at $\varphi = 0.325$ only.*

Figure 7 [written] placed. [fix] Caption for (a): mean of five packings and their range, not "a single seed". Add the arrow and the gray band. A full caption was drafted on 2026-10-11.

5. **Optional paragraph: what the scatter is.** (new) [to write]

   Message: *Differences between packings in the exponent are real properties of the packings, not noise from a small domain.*

   The scatter in the exponent does not fall with domain size while the scatter in the level does (11% to 4.5%), so the first belongs to the packing and the second to sampling. *Open: include here, in Section 5.3, or leave out.*

5 Discussion

Draft: 4 Discussion, treated as notes. Thirteen paragraphs. For each: the argument, the numbers that carry it, and what the notes offer about the literature.

Section opening (L412–421). [trim] Not needed; the subsections carry it.

5.1 What Vapor Transport Alone Reproduces

1. **The unsaturated pair** (L436–453). [written]

   Message: *The shortfall against Molaro et al. is not a tuning problem within our model: vapor transport through air cannot supply all of the measured growth. What supplies the rest is open; it is not surface diffusion by Molaro's own equations.*

   This is the one paragraph already written for the current results. It says: humidity inferred from shrinkage, $\alpha_c = 0.1$, under at $-20\,^{\circ}\mathrm{C}$ and over at $-5\,^{\circ}\mathrm{C}$, the authors' caveat, surface diffusion absent. Add:

   - The numbers, 41% and 170%.
   - That 41% is not a matter of tuning. With α_c at the top of its range the neck is insensitive to the chamber humidity, and with the vapor diffusivity raised 30 or 100 times the model still reaches only 75% of the measured growth once the grain shrinkage is matched; the two cases land 0.01 μm apart (tests of 2026-09-09). The vapor route has a ceiling below the measurement.
   - What fills the gap: not established. **Do not attribute it to surface diffusion.** Evaluating Molaro's own equations at the necks they measured (2026-10-13; Appendix A of this outline), surface diffusion is below 1% of their vapor term; it dominates in their model only for necks smaller than about 4 μm. Their model matches the data through its vapor term alone, which assumes $\alpha_c = 1$ and no gas diffusion. Say that the experiment grows the neck faster than vapor transport through air at $\alpha_c \leq 0.1$ allows, and name the candidates without choosing: transport faster than diffusion in still air (gradients or convection in the stage), an effective α_c above the literature range, surface diffusion with a coefficient far above published values.
   - A caution on the comparison itself: Molaro et al. present their own model's agreement as order-of-magnitude, and its vapor term assumes $\alpha_c = 1$ with no gas diffusion.
   - Why $-5\,^{\circ}\mathrm{C}$ is set aside: the authors' doubts about that series (illumination and gradient-driven sublimation, stronger near melting).

Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`; Swinkels & Ashby 1981 `swinkelsSecondReportSintering1981`; possibly Gundlach et al. 2018 `gundlachSinteringSublimationMicrometresized2018` on sintering of micrometre ice grains (in your `.bib`; check relevance).

2. **The saturated pair, and exponents.** (new) **[to write]**

   Message: *The model reproduces how the growth exponent depends on saturation. Where its exponent falls short of the measured one, the cause is the diffuse interface on a small neck, not a missing mechanism: the sharp-interface law for the same run gives 0.29.*

   - The comparison of exponents, as one small table or one sentence: this model at saturation 0.21–0.23 (phase field) and 0.29 (Gibbs–Thomson prediction for the same run); this model at Molaro's humidity, −20 °C, 0.135; Demmenie et al. 0.26–0.33 at saturation and about 1/7 undersaturated; Molaro's −20 °C data 0.23 ± 0.03; theory 1/3. No exponent is quoted at −5 °C.
   - What it supports: the exponent is set by the saturation state, as Demmenie et al. argue, and the model follows that trend.
   - Why the phase-field value at saturation is low: the evidence of Section 3.2, in two sentences. A diffuse interface cannot represent a fillet a few interface widths in radius; it fills it too fast.
   - Why not surface diffusion. Two reasons. Mechanisms acting in parallel add their rates, and surface diffusion alone gives $t^{1/7}$, so adding it could not raise an exponent toward 1/3. And by Molaro's own equations it is below 1% of vapor transport at necks of tens of micrometres (Appendix).
   - The consequence for the aggregates, stated plainly: their necks are also under-resolved early, which is why the first day depends on ε and why curves are normalized at $\theta_r = 30$.

   Open: which saturated run the paper quotes: the 100 h run from a pre-formed neck (0.21), or also the earlier run from tangent contact (0.283), which predates later solver changes and whose raw output is not on this machine.

   Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`; Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

3. α_c **as an effective parameter.** (new) **[to write]**

   Message: *The uncertain coefficient changes how fast things happen in the model, not the relation between conductivity and microstructure, so the main result does not rest on it.*

   The rates in this paper scale with $1/\alpha_c$; the relation between conductivity and SSA does not depend on it. Show the dependence (requested 2026-10-12):

$$\tau_{\mathrm{sub}} = \frac{\varepsilon^2}{d_0}\left(\beta_{\mathrm{sub}} + \Delta\beta\right), \qquad \beta_{\mathrm{sub}} = \frac{1}{\alpha_c}\,\frac{\rho_{\mathrm{ice}}}{\rho_{vs}^{I}(T)}\sqrt{\frac{2\pi m}{k_B T}},$$

   with $\Delta\beta$ independent of α_c. So $\tau_{\mathrm{sub}} \propto 1/\alpha_c$ as long as $\beta_{\mathrm{sub}} \gg \Delta\beta$, which holds in the aggregates ($\Delta\beta/\beta_{\mathrm{sub}} = 0.02$): a tenfold larger α_c would make every aggregate curve run ten times faster and leave it unchanged against θ or SSA. **[from the notes]** Bouvet et al. fit $\alpha_c = (9.8 \pm 0.7) \times 10^{-4}$ to tomography at −2 °C (L482–490); Braun et al. find 10^{-3}–10^{-1}, changing as surfaces facet, and call it an effective, surface-averaged value (L490–503); Kaempfer & Plapp's validation by bubble migration (L422–435).

Cite: Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Braun et al. 2024 `braunRigorousApproachSpecific2024`; Kaempfer & Plapp 2009 `kaempferPhasefieldModelingDry2009`; Granger et al. 2021 `grangerOrientationSelectiveGrain2021` for faceting under a gradient. The notes also cite Libbrecht 2013, 2017 `libbrechtMeasurementsSurfaceAttachment2013`, `libbrechtPhysicalDynamicsIce2017` for $\sigma_0(T)$: usable only as background on why α_c is uncertain.

5.2 Temperature as a Rescaling of Time

4. **Why the curves superpose.** (new) **[to write]**

   Message: *Explain the temperature result as a similarity argument: written in dimensionless form, the model's temperature dependence goes almost entirely into one time scale, so temperature can only rescale time; the simulations show how precisely.*

   Three steps (worked out 2026-10-12; full derivation with every step and the sizes of what is neglected in `temperature_scaling/time_temperature_scaling.pdf`):

   - Scale lengths by the grain radius $\bar{R}$, supersaturation by $d_0/\bar{R}$, and time by $\tau = \bar{R}^2\beta_{\mathrm{sub}}/d_0$ (proportional to τ_{sub}). The interface condition becomes parameter-free. The saturation vapor density, with its 31-fold change from -40 to $-5\,^\circ$C, the capillary length and $\sqrt{T}$ are all inside τ.
   - One temperature-dependent group is left, from the balance between vapor diffusing to a surface and ice attaching to it: $\bar{R}/L^*$ with

     $$L^* = \frac{D_v(T)}{\alpha_c}\sqrt{\frac{2\pi m}{k_B T}}.$$

     The saturation vapor density cancels in L^*; what remains is the vapor diffusivity. L^* is 123 μm at $-40\,^\circ$C and 147 μm at $-5\,^\circ$C (125 and 150 with the thin-interface term): it changes by 20% where τ changes by a factor of 29.
   - So both temperature-dependent properties are accounted for: the saturation vapor density sets the clock, and the diffusivity is a weak remaining correction. That correction is the likely source of the 4% by which rates depart from τ_{sub}, while the path in conductivity against SSA is unaffected to 0.27%. (Its size is consistent; that it is the cause has not been tested.)

   Say plainly that with one sintering mechanism this is close to a consequence of the construction; the numerical result is how well it holds with heat and vapor transport both active. Do not say gas diffusion is negligible: $\bar{R}/L^* \approx 0.4$, so it contributes; what is small is its *change* with temperature.

   Cite: none needed; if one, Kaempfer & Plapp 2009 for the attachment-limited limit.

5. **What would break it, and the activation energy.** (new) **[to write]**

   Message: *The collapse is therefore a signature of a single route. Its temperature dependence differs from the one measured for strength gain, which points to a second route in real samples and gives experimenters something to measure.*

   The clock's temperature dependence is an activation energy of about 50 kJ mol^{-1}, that of the saturation vapor density. **[from the notes]** Choukroun et al. measure 24.3 ± 3.3 kJ mol^{-1} for strength gain and attribute it to surface self-diffusion (L613–620), at 193–243 K; the two temperature ranges overlap at 233–243 K. The difference is consistent with a second route in their samples, and an activation energy is something a laboratory can measure to tell the

routes apart. A temperature-dependent α_c, a second route, or a temperature gradient would each break the superposition, which makes it a test.

Cite: Choukroun et al. 2020 `choukroun_strength_2020`.

5.3 Specific Surface Area as the State Variable

6. **Density is not enough.** (new) **[to write]**

   Message: *Conductivity changes while porosity stays fixed, so a density-based formula cannot describe a sintering deposit.*

   In these runs porosity is fixed and conductivity still rises, by 17% between $\theta = 30$ and 331. **[from the notes]** Calonne et al. (2011): tensor from 30 tomographic images, a regression on density, scatter attributed to anisotropy (L507–517). Calonne et al. (2014): under a temperature gradient conductivity changes at constant density, with SSA and anisotropy (L517–529). Half the length of the notes.

   Cite: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`; Calonne et al. 2014 `calonneStudyTemperatureGradient2014`; Schneebeli & Sokratov 2004 `schneebeliTomographyTemperatureGradient2004` (the notes cite it for the same behavior under a gradient).

7. **The power law and where it stops.** (new) **[to write]**

   Message: *State the law, its range, and that outside the range we make no claim.*

   $p = -0.80$ for $\varphi \leq 0.375$; the level from porosity; above 0.375 the exponent weakens and the packings are anisotropic. No claim that packings of different porosity converge.

   Cite: Calonne et al. 2011, 2019 `calonneNumericalExperimentalInvestigations2011`, `calonneThermalConductivitySnow2019` for conductivity against density, as the contrast.

8. **Why packings differ.** (new) **[to write]**

   Message: *Packings of the same porosity conduct differently because their ice networks are arranged differently; at high porosity the networks are anisotropic and the simple law fails in one direction.*

   Packings of one porosity differ in how well the ice is connected, and the exponent follows that (correlation -0.77 with conduction efficiency), not coordination number or anisotropy. In these packings the ice always spans the cell and the pore never does (percolation test, 2026-10-12): the pore space is a set of closed pockets, which is what "sealed pores" means here, and in two dimensions only one phase can span. What differs between packings is the quality of the ice network, not whether it exists: at high porosity the packings are anisotropic, grains dropped under gravity being supported from below with fewer contacts sideways, and the power law survives along the deposition direction while failing across it. Absolute values are therefore not compared with three-dimensional measurements, where both phases are connected. *Open: literature thresholds for discs are not yet checked against a source; whether to show the efficiency; the Euler-characteristic series (tested on three runs) is available if a direct measure is wanted.*

5.4 What a Laboratory Could Measure

9. **Three tests.** (new) **[to write]**

   Message: *Turn the results into measurements someone could make on sintered ice-sphere samples.*

On unloaded ice-sphere samples held isothermally: (i) curves at different temperatures superpose under a time shift, and the shift gives an activation energy; (ii) conductivity against SSA is one curve; (iii) SSA falls linearly in log time. **[from the notes]** SSA is measured by near-infrared reflectance, gas adsorption and tomography (L562–568; Gallet, Matzl, Proksch).

Cite: Gallet et al. 2009 `galletMeasurementSpecificSurface2009`; Matzl & Schneebeli 2006 `matzlMeasuringSpecificSurface2006`; Proksch et al. 2015 `prokschDensitySpecificSurface2015`; for isothermal SSA decay in the laboratory, Legagneux & Dominé 2005, Taillandier et al. 2007, Kaempfer & Schneebeli 2007 `kaempferObservationIsothermalMetamorphism2007`, Löwe et al. 2011 `loweInterfacialStructuralRelaxations2011` (forms only).

5.5 Implications for Plume Deposits on Enceladus

10. **What carries over.** (new) **[to write]**

    Message: *For Enceladus the qualitative conclusion holds: a sintering deposit conducts better as its surface area falls, even with no compaction.*

    Qualitative: as a deposit sinters, SSA falls and conductivity rises together, without densification. **[from the notes]** Tiger Stripe temperatures up to about 180 K (Spencer); grain size increasing toward the fractures in VIMS data (Jaumann; L580–602); Molaro et al. on the dependence of sintering on temperature and grain size.

    Cite: Spencer et al. 2009 `spencerEnceladusActiveCryovolcanic2009`; Spencer & Nimmo 2013 `spencerEnceladusActiveIce2013`; Jaumann et al. 2008 `jaumannDistributionIcyParticles2008`; Molaro et al. 2019; Choukroun et al. 2020.

11. **Why it matters, and the gap.** (new) **[to write]**

    Message: *Conductivity of the surface layer affects thermal and radar interpretation, but our conditions are far from the moon's, so we give no timescales.*

    [from the notes] A low-conductivity surface layer changes subsurface temperatures and radar attenuation (Byrne; L603–612); Choukroun et al. on strength gain and fallout rates of 0.01–1 $\mathrm{mm\,yr^{-1}}$ (L613–629). Then the gap, stated plainly: the runs stop at $-40\,^{\circ}\mathrm{C}$ and use 50 μm grains in air; the moon is far colder, the grains are micron-sized, and there is no air. No timescales are given.

    Cite: Byrne et al. 2024 `byrneRadarAttenuationSubsurface2024` (a preprint DOI in the `.bib`; check whether it is published); Choukroun et al. 2020; possibly Ferrari et al. 2021 `ferrariThermalEmissionSaturns2021` on thermal emission of Saturn's icy moons (in your `.bib`; check relevance).

5.6 Limitations

12. **Model.** (new) **[to write]**

    Message: *What physics is missing.*

    Vapor route only; constant α_c; air-filled, sealed pores at 1 atm; conduction only and constant k_{ice}, k_{air}. **[from the notes]** Fourteau et al. on the contribution of vapor transport to apparent conductivity under a gradient (L569–575; check the figure quoted); Calonne et al. (2019) on temperature-dependent conductivities (L575–579).

    Cite: Fourteau et al. 2021 — your `.bib` has two, `fourteauImpactWaterVapor2021` and `fourteauMacroscopicWaterVapor2021`; the point about apparent conductivity is the first. Calonne et al. 2019 `calonneThermalConductivitySnow2019`.

13. **Numerics and geometry.** (new) [to write]

    Message: *What the simulations cannot resolve or represent.*

    Two dimensions, so relative changes are claimed and absolute conductivities are not; one grain size distribution; the first day depends on the interface parameter, because a neck whose fillet is narrower than about 7 interface bands grows faster than the Gibbs–Thomson law (Section 3.2), so exponents fitted over that range are biased low; pore throats narrower than the diffuse band (9.2 μm) are not resolved, which raises k_{eff} most at low porosity; the power law is established for $\varphi \leq 0.375$; rates scale with $1/\alpha_c$, whose value is uncertain over two decades.

    Cite: none needed beyond those above.

In the notes and not to be carried over: $\beta_{\text{sub}}(T)$ from Libbrecht's $\sigma_0(T)$, a single tuning and a "genuine prediction" at $-5\,^{\circ}\text{C}$ (L454–481); isotropy, and a collapse across porosity (L530–561); sintering "on the order of decades" (L596) and conductivity starting "near 0.0" (L611).

6 Conclusions

Draft: 5 Conclusion, treated as notes.

1. **What was done and found.** (new) [to write]

   Message: *Three results, each with the range over which it was shown.*

   One sentence of method, then the three results with their ranges: temperature rescales time (-40 to $-5\,^{\circ}\text{C}$, within 0.3%); conductivity follows $\text{SSA}^{-0.80}$ with the level set by porosity ($\varphi \leq 0.375$); vapor transport gives 41% of measured neck growth at $-20\,^{\circ}\text{C}$.

2. **What comes next.** (new) [to write]

   Message: *The experiment and the model extensions that would test or extend them.*

   The laboratory test of Section 5.4; a second transport route in the model; three dimensions; colder temperatures and smaller grains.

Open Research; Acknowledgments

[to write] Code and data availability; run inventory (`RUN_TABLE.md`). Acknowledgments.

Supporting Information

- **Figures.** S1 domain-size convergence; S2 interface parameter; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Tables, text.** Complete parameters, if moved; packing construction; the weak-form derivation, if trimmed; run table.

Appendix: the two calculations behind Sections 3 and 5.1

A. Surface diffusion against vapor transport, by Molaro's own equations

Question. Does surface diffusion supply the 59% of neck growth our model lacks at −20 °C?

Equations (Molaro et al., 2019, Table 3 and Eqs. 1–4, 6, 17–20; a grain radius, x neck radius, S their smoothing factor):

$$\dot{V}_1 = S\,\frac{3\pi x\,\delta_s D_s\,\gamma_s\Omega\,(K_3 - K_2)}{d_2\,kT} \qquad \text{surface diffusion}$$

$$\dot{V}_3 = S\,2\pi x\rho\theta\,P_v\,\frac{\gamma_s\Omega}{kT}\left[\frac{\Omega}{2\pi\Delta_o kT}\right]^{1/2}(K_3 - K_m) \qquad \text{vapor transport}$$

$$\dot{x} = \frac{1}{2\pi x\theta\rho}\sum_i \dot{V}_i$$

$$K_3 = \frac{2}{a},\quad K_m = -\frac{1}{\rho} + \frac{1}{x},\quad \rho = \frac{x^2}{2(a-x)},\quad \theta = \tan^{-1}\frac{a}{x+\rho},\quad K_1K_2 = K_m^2,$$

$$\frac{K_2 - K_1}{K_1} = \frac{3A\rho}{x} + \frac{3}{2} - \frac{3}{2}\left[3\Big(\frac{A\rho}{x}\Big)^2 + \frac{4A\rho}{x} + 1\right]^{1/2},\quad d_2 = \frac{\rho}{1 + \left(\frac{4(K_2-K_1)}{3K_2}\right)^{1/2}},\quad A = \frac{\delta_b D_b}{\delta_s D_s}.$$

The vapor term is Hertz–Knudsen with $\alpha = 1$ and no gas diffusion (their Appendix A).

Parameters (their Table 5): $\Omega = 1.181\times10^{-29}$ m^3; $\Delta_o = 917$ kg m^{-3}; $\gamma_s = 0.06$ J m^{-2}; $\delta_s = \delta_b = 9.04\times10^{-10}$ m; $D_{0s} = 1.4\times10^{-8}$ m^2 s^{-1} (literature range down to 1.35×10^{-13}), $Q_s = 23.2$ kJ mol^{-1}; $D_{0b} = 8.4\times10^{-4}$ m^2 s^{-1}, $Q_b = 4.9$ kJ mol^{-1}; P_v from Murphy & Koop (2005), 103 Pa at −20 °C.

At their experiment ($a = 87$ μm, −20 °C; measured mean $\dot{x} = 3.4\times10^{-9}$ m s^{-1}):

x [μm]	ρ [μm]	$\dot{V}_1/\dot{V}_3$	$\dot{x}$ from $\dot{V}_1$ [m/s]	$\dot{x}$ from $\dot{V}_3$ [m/s]	$\dot{V}_1/\dot{V}_3$, $A = 0$
16.4	1.9	6.7×10^{-3}	5×10^{-11}	7.8×10^{-9}	8×10^{-7}
20.0	3.0	3.1×10^{-3}	2×10^{-11}	4.9×10^{-9}	3×10^{-7}
25.0	5.0	1.3×10^{-3}	4×10^{-12}	2.9×10^{-9}	1×10^{-7}
32.4	9.6	4.1×10^{-4}	6×10^{-13}	1.5×10^{-9}	4×10^{-8}

- Surface diffusion is 0.7% falling to 0.04% of their vapor term. It exceeds it only for x below about 4 μm.
- Their vapor term alone gives the measured rate.
- The last column switches off the coupling to grain-boundary diffusion ($K_2 = K_m$). With their parameters $A = 3.5\times10^8$, and the coupling raises the driving curvature K_2 by a factor of about 4000.

Check and its limit. Their Figure 10 (25 μm, $T/T_m = 0.8$) was recomputed. The vapor current agrees (2×10^{-21} rising to 3×10^{-20} m^3 s^{-1}; theirs about 10^{-21} to 2×10^{-20}). The surface current starts about 100 times above theirs, because they solve the four coupled equations iteratively where this uses the approximate Eq. (20). The table therefore *overstates* surface diffusion. Script and output: `studies/molaro_2019/sa81_mechanisms/`.

What it implies. The gap is between our vapor route ($\alpha_c = 0.1$, diffusion through air) and whatever supplies vapor in the experiment. A one-equation neck model fitted to the same data wants $\alpha_c \approx 0.3$–1 (`studies/molaro_2019/envelope_ode/`).

B. The saturated pair against the Gibbs–Thomson law

Question. Is the exponent of 0.21 physics or resolution?

Prediction. For a neck of radius x between grains of radius R, with fillet radius $\rho = x^2/(2(R-x))$, fed from the grain surface through attachment in series with gas diffusion over a length $c\rho$:

$$v_n = \frac{d_0 \left(\dfrac{1}{\rho} - \dfrac{1}{x} + \dfrac{2}{R} \right)}{\beta_{\mathrm{sub}} + c\,\rho\,k_D}, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\pi}{2}\, v_n, \qquad k_D = \frac{\rho_{\mathrm{ice}}}{D_v\, \rho_{vs}^I}.$$

Attachment alone gives $x^3 \propto t$; diffusion alone $x^5 \propto t$. $R = 84.4$ μm, -20 °C, $\alpha_c = 10^{-3}$; $c = 0.275$ from the Molaro run, not refitted. Here gas diffusion is 0.4–2% of the resistance. Integrated from the run's first sample after relaxation, it gives a final width of 60.6 μm (run: 63.2) and an exponent of 0.29 (run: 0.21).

Run against prediction ($\varepsilon = 0.12$ μm; one interface band $= 9.19\,\varepsilon = 1.1$ μm; exponents from fits of $C(t + t_0)^a$ starting at the time shown):

Start	w [μm]	w/ε	fillet [bands]	velocity, run / prediction	a, run	a, prediction
2 h	35.1	293	2.1	2.54	0.215	0.292
6 h	39.2	327	2.7	1.72	0.221	0.292
16 h	44.9	374	3.7	1.36	0.227	0.290
31 h	50.1	418	4.8	1.18	0.231	0.288
51 h	55.0	459	6.0	1.08	0.234	0.287
71 h	58.8	490	7.1	1.02	0.236	0.286

The excess fits $1 + 11.4\,(\mathrm{band}/\mathrm{fillet})^{2.7}$ (rms 0.035).

Sensitivity to the assumed contact time (same data, t_0 fixed):

t_0	4.8 h (free fit)	8 h	12 h	16 h	24 h
exponent	0.212	0.234	0.258	0.281	0.325
rms misfit	0.28%	0.78%	1.33%	1.73%	2.31%

The prediction takes 16 h to grow a neck to the run's first fitted width.

What it shows and what it does not. The rate converges on the sharp-interface law as the neck becomes resolved, and the early excess accounts for most of the difference in exponent. It does not show the predicted exponent being reached (0.236 at the end against 0.29), and an early excess caused by the starting shape of the neck would look the same. A run at $\varepsilon/2$ would separate the two. Scripts and output: `studies/molaro_2019/demmenie/` (`predict_neck_growth.py`, `interface_width_estimate.py`).

Citations: what the .bib lacks

Suggested above and present in the `.bib`: all of them. Mentioned in this plan and *not* in the `.bib`:

- Auriault, Boutin & Geindreau (2009), the book whose naming of **t** the paper follows (doi:10.1002/9780470612033
- Ettrich et al. (2014), the tensor law applied to heat conduction; only if a second reference for the law is wanted.

- A source for packing and percolation thresholds of discs, if Section 5.3 states them.

Two keys are still in the old style in the `.bib` itself: `choukroun_strength_2020` and `nicoli_tensorial_2011`.

Smaller corrections throughout

- **Notation**, to the conventions listed at the top. Subscripts "ice" and "air" everywhere (ρ_{ice}, not ρ_i; ϕ_{ice}, ϕ_{air}); $\mathbf{K}_{\text{eff}}$ for the tensor and k_{eff} for its scalar mean, each used consistently; k_{ice}, k_{air} as in the tables; $\rho_{\text{air}}, P_{\text{atm}}$ in Eq. (2). ϕ_{air} is air, not water vapor.
- **Temperature.** Celsius with $T + T_0$ in Eqs. (3), (5); kelvin in Table 1. One convention.
- **Numbers.** 30 days (L172 says 28); mesh 2829×2829 (L362).
- **Words.** "Allen–Kahn"; "tonsorial"; "effect" for "affect"; "table table", "figure figure", "section section".